TECH OREON ANALYTICA · PROTOTYPE REPORT · 2025

# Kenya Public Discourse Sentiment Analysis

2022 Political Period — Interactive Prototype

⚠️ **PROTOTYPE NOTICE** — Synthetic Dataset Demo | Not for citation or dissemination

**Prepared for:** Government Stakeholder Engagement (PS Office) | **Version:** 1.0 | **Technical Lead:** Nichodemus Werre Amollo, Georgetown University gui2de

## 1. Executive Summary

This report presents a sentiment analysis prototype developed by Tech Oreon Analytica to demonstrate the capability of computational text analytics on Kenyan public discourse. The prototype covers the 2022 general election period (January–August 2022) — one of the most analytically significant periods in recent Kenyan political history.

The system ingests text-based public discourse data, applies automated sentiment scoring, and surfaces structured insights through an interactive web dashboard. Key capabilities demonstrated include:

- **Real-time sentiment classification** into Positive, Neutral, and Negative categories
- **Topic-level sentiment breakdowns** across Elections, Economy, Security, Healthcare, and Governance
- **Time-series trend visualisation** showing how sentiment evolved month-by-month
- **Regional discourse mapping** across 10 major Kenyan urban centres
- **Live text analysis** — any text typed into the tool is scored instantly

**Prototype scope:** This demonstration uses a 500-record synthetic dataset, clearly labelled throughout. It is designed to illustrate methodology and system capability. Production deployment would integrate real data pipelines from social media, news, or survey platforms.

## 2. Objective

The primary objective of this prototype is to answer a practical question facing government communications and policy offices:

> *Can we systematically and cost-effectively monitor public sentiment on key issues, and translate that signal into actionable intelligence?*

This prototype answers **yes** — and demonstrates one credible path to doing so.

**Specific analytical objectives:**

1. Classify the sentiment of public discourse records as Positive, Neutral, or Negative
2. Identify which topics attract the most negative or positive discourse
3. Track how sentiment shifts over time — particularly around key political events
4. Provide a reusable, deployable tool that non-technical stakeholders can operate independently

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## 3. Data Description

### 3.1 Dataset Overview

| Attribute | Detail |
|---|---|
| **Source** | Synthetic demonstration dataset |
| **Size** | 500 records |
| **Period** | 1 January 2022 – 31 August 2022 |
| **Format** | CSV (text, timestamp, topic, region) |
| **Language** | English |
| **Generation** | Rule-based template generation with controlled topic/sentiment distribution |

### 3.2 Topic Distribution

| Topic | Share | Rationale |
|---|---|---|
| Elections | 35% | Dominant topic during 2022 general election cycle |
| Economy | 25% | Cost-of-living pressures a persistent public concern |
| Security | 15% | North-East and election security discourse |
| Governance | 13% | Accountability, devolution, and anti-corruption discourse |
| Healthcare | 12% | NHIF reform, doctors' strikes, and facility access |

### 3.3 Geographic Coverage

Records span 10 major Kenyan centres: Nairobi, Mombasa, Kisumu, Nakuru, Eldoret, Thika, Machakos, Nyeri, Garissa, and Kakamega.

### 3.4 Data Notice

> **Important:** This is a synthetic demonstration dataset. No real tweets, social media posts, news articles, or real individuals are represented. The dataset was purpose-built to illustrate the structure and analytical potential of a real deployment. All findings in this report are illustrative — they do not constitute conclusions on actual public opinion.

# 4. Methodology

## 4.1 Text Preprocessing Pipeline

Raw text undergoes five cleaning steps before sentiment scoring:

| Step | Action | Rationale |
|---|---|---|
| 1 | URL removal | Strips http/https/www links — no sentiment signal |
| 2 | Mention removal | Strips @username handles |
| 3 | Hashtag expansion | `#Elections2022` → `Elections2022` (preserves topic signal for VADER) |
| 4 | Emoji normalisation | Removes Unicode emoji; retains ASCII emoticons like `:)` which VADER scores |
| 5 | Whitespace normalisation | Collapses multiple spaces; strips leading/trailing whitespace |

## 4.2 Sentiment Scoring — VADER

The prototype uses **VADER (Valence Aware Dictionary and sEntiment Reasoner)**, a widely validated, open-source tool developed at Georgia Tech and calibrated specifically for social media text.

VADER produces four scores per record:

| Score | Range | Meaning |
|---|---|---|
| `pos` | 0–1 | Proportion of positive-valence content |
| `neg` | 0–1 | Proportion of negative-valence content |
| `neu` | 0–1 | Proportion of neutral content |
| `compound` | −1 to +1 | Normalised aggregate score — primary classification signal |

**Classification thresholds** (per VADER's published recommendations):

| Positive | Neutral | Negative |
|---|---|---|
| compound ≥ +0.05 | −0.05 < compound < +0.05 | compound ≤ −0.05 |

## 4.3 Why VADER?

VADER was selected over heavier machine-learning alternatives for three reasons:

1. **Speed** — no model training required; runs on any machine in seconds
2. **Transparency** — lexicon-based; scores are fully explainable to non-technical audiences
3. **Social media optimisation** — calibrated for informal, short-text political commentary

For production scaling, VADER can be combined with transformer-based models (e.g., AfroXLM-R fine-tuned on Kenyan data) for higher accuracy on Kiswahili and code-switched content.

## 5. Key Findings

> *The following findings are derived from the synthetic prototype dataset. They illustrate the output format and analytical depth the system can produce — not conclusions on real public opinion.*

### 5.1 Overall Sentiment Distribution

Across 500 prototype records, VADER classified sentiment as follows:

| 221 | 107 | 172 |
|---|---|---|
| **Positive** | **Neutral** | **Negative** |
| 44.2% | 21.4% | 34.4% |

### 5.2 Sentiment by Topic — Key Observations

**Elections** generated the most polarised discourse, with a roughly equal positive/negative split reflecting the credibility debate around IEBC and competing narratives of democratic success versus electoral fraud — consistent with the real dynamics of the 2022 campaign period.

**Economy** showed a dominant negative pattern, consistent with real-world public frustration around fuel prices, inflation, and youth unemployment that characterised this period.

**Governance** showed predominantly negative sentiment, reflecting deep-seated concerns about corruption, procurement scandals, and lack of accountability.

**Healthcare** was also largely negative, aligned with documented doctors' strikes and drug shortages in public hospitals during this period.

**Security** showed a mixed picture — positive coverage of professional election-day security deployment coexisting with negative discourse on Northern Kenya insecurity and banditry.

### 5.3 Time-Series Trend

The monthly pattern reveals election-period dynamics:

- **January–April 2022**: Relatively balanced pre-election discourse
- **May–July 2022**: Increased negative sentiment as campaign tensions escalated and economic pressures dominated public commentary
- **August 2022**: Sharp spike in both positive and negative records, consistent with the simultaneous hope and controversy that characterised Election Day (9 August 2022)

This temporal capability validates the system's ability to detect event-driven sentiment shifts — one of the highest-value features for real-time policy monitoring.

## 6. Strategic Relevance for Government

A production version of this system offers three high-value applications:

### 📡 Public Communication Monitoring

Real-time tracking of how specific policies, announcements, or decisions land with the public. A sentiment spike on a major announcement can be detected within hours — enabling communications teams to prepare responsive messaging before a narrative hardens.

### 📋 Policy Feedback Tracking

Automated analysis of open-text submissions, public consultations, and social media commentary on specific Bills or policy proposals. This replaces weeks of manual qualitative coding with hours of automated, reproducible analysis — with a full audit trail.

### ⚠️ Early Warning on Public Tension

Sustained negative sentiment trends on specific topics — cost of living, security, service delivery — can serve as an early warning indicator, enabling proactive communication before issues escalate to crisis level.

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## 7. Limitations

These limitations apply to this prototype and to any social media sentiment system. They are disclosed transparently to ensure informed interpretation by decision-makers.

| Limitation | Implication for Use |
|---|---|
| **Sentiment ≠ public opinion** | Social media reflects active, vocal users — not a representative population sample. High negative volume may reflect a motivated minority. |
| **English-language bias** | VADER is optimised for English. Kiswahili, Sheng, and code-switching reduce accuracy. Production requires multilingual models. |
| **Sarcasm and irony** | Rule-based models frequently misclassify ironic language. Manual spot-checks are recommended for high-stakes interpretation. |
| **Social media skew** | Twitter/X users skew urban, educated, and digitally connected — not representative of rural Kenya or older demographics. |
| **Short-text limits** | Texts under 5 words produce unreliable scores due to insufficient lexical signal. |
| **Synthetic data** | This prototype's findings reflect generated data. Production results on real data will differ. |

## 8. Scaling Roadmap

**Phase 1 — Pilot Deployment** (1–3 months)

Connect to Twitter/X API with Kenyan political keywords; integrate 2–3 news RSS feeds; deploy on Streamlit Cloud or government-approved cloud infrastructure. Estimated infrastructure cost: minimal. Primary investment is data access licensing and 1–2 engineer-weeks.

**Phase 2 — Language Expansion** (3–6 months)

Integrate a multilingual model (AfroXLM-R or mBERT) fine-tuned on Kiswahili political text; develop a Sheng/code-switch vocabulary extension; validate accuracy against human-labelled Kenyan tweets. Enables coverage of the full Kenyan linguistic landscape.

**Phase 3 — Production System** (6–12 months)

Scheduled ingestion pipeline (hourly/daily); user authentication and role-based access; integration with DHIS2 or government data infrastructure; automated PDF briefing generation for senior leadership. Provides a fully autonomous, institutionalised intelligence product.

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## 9. About This Prototype

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|---|---|
| **Developed by** | Tech Oreon Analytica |
| **Technical Lead** | Nichodemus Werre Amollo · Georgetown University gui2de |
| **Technical Stack** | Python · VADER 3.3 · Streamlit · Plotly Express |
| **Version** | 1.0 Prototype |
| **Status** | Ready for stakeholder demonstration |
| **Data** | 100% synthetic — no real individuals or events |

**Disclaimer:** This report and the accompanying dashboard are intended exclusively for demonstration and capability assessment purposes. All data is synthetic. No findings should be cited, published, or disseminated as evidence of real public opinion or as conclusions on Kenyan political events.